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One of the main concerns that NASA has about the long-term health of the astronauts who go to Mars is the impact that cosmic radiation will have on brain function. Cosmic radiation consists of charged particles, known as Hze particles. The biological effects of Hze particles are dependent upon how much energy the particles deposit when they interact with tissue, which is referred to as LET. At present we only have superficial knowledge of how some high LET radiations inhibit brain function. Our current data suggest that brain function is inhibited due to radiation-induced death of brain cells and the inhibition of the brain's signaling pathways. However, the relative importance of each of these processes at different LETs is unclear. Such information is not only vital to the development of risk assessment models, but may provide an opportunity to develop medical treatments that can slow down or even stop the progression of radiation-induced brain impairment.
The central objectives of our proposed studies address Research questions 29e, 29h and 29i outlined in the current Bioastronautic Roadmap. We specifically propose to:

1) identify the high-LET radiations that have the greatest impact on the ability of animals to recall imprinted memories and to learn new memory skills;

2) use PET imaging to identify the regions of the brain that are functionally inactivated when brain function is inhibited by Hze radiation

3) establish the relative importance of radiation-induced cell death and signal malfunction in the inhibition of brain function after exposure to radiations with low, medium and high-LETs;

4) develop a blood-based assay that can be used to monitor the astronauts for early signs of radiation-induced memory function deficit.

Task Description:

Specific Aim 1 will determine the dose required to inhibit four different memory functions in rats exposed to radiations with different LETs. The heads of rats will be exposed to graded doses of various LET radiations and the rats monitored over the following 6 months for signs of an impairment of their ability to retrieve memory and to learn new memory dependent tasks.

Specific Aim 3 will determine the changes in the brain that lead to the loss of memory function in rats exposed to high-LET radiations. We shall determine the impact that radiation exposure has on the proliferation of brain cells by microscopic examination of brain slices. We shall also determine if radiation affects the ability of brain cells to release signaling proteins in response to known stimuli. Finally we shall use the new technology of proteomics, which can simultaneously measure the abundance of thousands of proteins, to identify all processes that are altered by radiation and are associated with the loss of brain function.

Specific Aim 4 will use proteomics to establish the serum protein profiles of rats (who receive whole body radiation exposures) that go on to develop radiation-induced impairment of brain function. These proteins can be used as biomarkers with which to identify astronauts who have been exposed to radiation doses/energies that put them at risk of developing brain impairment.

Our studies will establish the safe dose of cosmic radiation that astronauts can be exposed to without suffering from brain malfunction, which will be invaluable in deciding if the Mars mission should proceed, and if so, what additional precautions will be needed to protect the astronauts. Furthermore, our studies will provide considerable insight into how high LET radiation exposure leads to brain function impairment. These data will help develop treatments that can reduce the severity of radiation-induced brain impairment, e.g., special diets to reduce radiation-induced brain cell death, or drugs that can alter the brain's signaling pathways. Finally, our studies will identify biomarkers that can be used to identify and monitor astronauts who are at risk of developing radiation-induced brain impairment.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The identification of serum biomarkers for cognitive impairments has potential benefits in the diagnosis of neurological disorders.

During this reporting period we have primarily focused on two areas:
1. Determining the dose of Hze radiation that leads to significant impairment in neurocognitive function.

2. Identifying Hze-induced changes in neuropeptide release and the CNS proteome that are associated with neurocognitive impairment.

Our most notable progress during this reporting period is that we have determined that exposure to Hze (1 GeV 56Fe) radiation doses that the astronauts on the Mars mission are likely to receive (i.e. 20 cGy) results in significant impairment of two neurocognitive functions in our rat model system. The ability of rats exposed to 20 cGy Hze to perform spatial memory and executive function (attentional set shifting) tasks was significantly impaired when assessed at 3 months post exposure. Preliminary data suggests that the Hze-induced impairment of these neurocognitive tasks may be partially reversible, given that we observe no significant impairment in spatial memory and executive function 6 months after exposure to 40 cGy. Over the remainder of the grant we shall determine if lower radiation doses also result in significant impairments of these neurocognitive tasks, and whether there is any recovery of this impairment.

Our observation that low (20 cGy) Hze doses result in significant impairment of neurocognitive function raises the possibility that radiation-induced cell death of neural cells and/or the impairment of neurogenesis may not be the primary cause of Hze-induced neurocognitive function. Our working hypothesis is that Hze-induced neurocognitive impairment arises through changes in synaptic plasticity, and thus we have also determined the impact that Hze exposure has on the neuropeptide release machinery within the brain. During this reporting period we have determined that 60 cGy of 1 GeV 56Fe radiation significantly ($p<0.05$) reduced hyperosmotic sucrose evoked glutamate release, a measure of the readily releasable vesicular pool (RRP). This HZE-induced reduction in the glutamatergic RRP persisted for at least six months post exposure. HZE radiation also significantly reduced levels of NR1, NR2A, and NR2B, subunits of the glutamatergic NMDA receptor, which is critical to synaptic plasticity, three months post irradiation. NR2A level remained suppressed at six months post exposure, but the level of NR2B and NR1 receptors returned to or exceeded normal levels respectively. Overall, this study shows that hippocampal glutamatergic transmission is sensitive to relatively low doses of 56Fe particles. The observed functional impairment of glutamatergic transmission, at both the

Task Progress:	pre- (RRP) and post- (NMDA receptor) synaptic level is likely to have a significant effect on learning and memory, and offers an explanation for the loss of neurocognitive function after exposure to Hze doses that do not lead to any significant changes in neurogenesis. Effective spatial memory is thought to be primarily dependent upon a fully functional hippocampus. We have established that Hze exposure leads to changes in neurotransmission in the hippocampus, and we have used standard investigational steps to establish some of the proteins whose expression changes after Hze exposure. However, we are cognizant that there are thousands of proteins and possibly hundreds of pathways that could be altered within the hippocampus that could cause the impairment of spatial memory. Using traditional focused approaches would require several years and thousands of dollars to fully establish what changes in the hippocampal proteome may be the root cause of the Hze-induced spatial memory impairment. Our center is a world leader in proteomic profiling for several disease states, and we have started to apply this expertise to conduct an unbiased investigation of the changes in the proteome of the hippocampus that correlate with good or bad spatial memory performance following Hze exposure. During this reporting period we have established in our laboratory a technique that uses trypsin-catalyzed incorporation of 18O into peptides that facilitates a semi-quantitative analysis of differentially expressed peptides in hippocampal protein extracts. Our first experiments have shown that there are detectable differences in the hippocampal protein composition of rats that have good versus bad spatial memory. Neurocognition is a complex process that often requires input from, and interactions between, multiple brain areas. Radiation-induced cognitive failure could thus occur through direct damage of one critical brain region, or to the ancillary brain regions that might be considered the “fine tuners”. We have thus conducted some parallel experiments using a relatively unique technique called MALDI-IMS. MALDI-IMS tissue profiling has the power to link the molecular detail of mass spectrometry with molecular histology, generating mass spectra from defined locations within a 20 um tissue section. We have used MALDI-IMS to determine the Hze-induced changes in the proteomic profile of the rat’s brain with a high degree of spatial resolution. Briefly, a region of the brain was selected that contains multiple regions of interest (ROI) in the hippocampus and cortex, using “The Rat Brain in Stereotaxic co-ordinates” (Paxanos & Watson, Elsevier). Using the appropriate co-ordinates we use a cryostat to obtain tissue slices from this region. Adjacent slices are processed with conventional H&E staining, or processed for MALDI-MSI. The entire brain section is progressively scanned by a smart beam laser and the mass-spectra profile obtained at each raster position is stored. Subsequently we define a ROI, based upon the co-ordinates of histological features on the H&E stained section, and then recover the peptide profile from all the raster positions within that gated region. The composite data can then be viewed as a “gel-view” of the peptide profile from the ROI, such as the hippocampus, and proteins whose expression changes in accordance with Hze-induced spatial memory identified. Thus MALDI-MSI can be used to gain insight into protein changes within a specific region that may be the underlying cause of the spatial memory impairment. It is also possible to use this technique to identify regions of the brain which have not previously been thought to impact upon a particular neurocognitive function. It is possible to take the individual mass spectra from each raster point and construct a 2D “heat-map” that shows the spatial distribution of specific peptides within the entire brain. Such heat-maps allow for the visual identification of regions of the brain that show proteomic differences that correspond to spatial memory impairment. Using this technique we have found that a peptide with a m/z of 14207 is differentially elevated in the Thalamus of irradiated rats that have good spatial memory. MALDI-MSI thus appears to be a powerful tool that can be used to identify radiation-induced changes in ancillary brain regions that correlate with neurocognitive impairment. Over the next year of the grant, we shall continue to use MALDI-MSI to identify region-specific proteomic changes that correlate with an individual rat’s cognitive performance status.

Bibliography Type:	Description: (Last Updated: 03/31/2026)
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